Networks and Complexity

Solution 15-4

This is an example solution from the forthcoming book Networks and Complexity. Find more exercises at https://github.com/NC-Book/NCB

Ex 15.4: aSIS moment expansion [Lvl. 2]

In the aSIS model infected nodes recover at the rate r, the infection spreads across SI links at rate p and susceptible nodes rewire their SI links from infected neighbors to randomly chosen susceptible partners at rate w, derive a differential equation for the density of SS-links, $[SS]$, as a function of the density of SI-links, $[SI]$, and SSI-triplet chains, $[SSI]$ (i.e. derive $[\dot{SS}] = r[SI] - p[SSI] + w[SI]$.)

Solution

To derive the differential equation, we consider the processes one-by-one. The easiest one is rewiring. When an SI-link is rewired by the susceptible node it turns into an SS-link. So this process produces SS-links at the rate

$$wK_{\mathrm{SI}}. \tag{1}$$

Next we consider recovery. Like rewiring, recovery can only ever be a source of SS-links. If a node recovers, it creates a new SS-link for every neighbor in state S that the recovering node has. Because our focal node is in state I before recovery, the links that are turned into SS-links upon recovery are SI-links before. So, to estimate the number of SS-links created in a typical recovery event we need to estimate the average number of SI-links on an I-node.

The average number of SI-links per I-node is K_{SI}/I. Multiplying the rate of recovery events, rI, we find the production rate of SS-links from recovery to be

$$rI\frac{K_{\mathrm{SI}}}{I} = rK_{\mathrm{SI}}. \tag{2}$$

Finally, we consider infection, which can only appear as a sink of SS-links. When an infection event happens, the number of SS-links that are destroyed is identical to the number of SSI-triplet chains running through the SI-link on which the infection occurs, which is $T_{\mathrm{SSI}}/K_{\mathrm{SI}}$. Multiplying the rate of infection events, pK_{SI}, we find the total rate of destruction of SS-links to be

$$pK_{\mathrm{SI}}\frac{T_{\mathrm{SSI}}}{K_{\mathrm{SI}}} = pT_{\mathrm{SSI}} \tag{3}$$

Taking all of the terms into consideration, we arrive at

$$\dot{K}_{\mathrm{SS}} = rK_{\mathrm{SI}} - pT_{\mathrm{SSI}} + wK_{\mathrm{SI}} \tag{4}$$

Dividing everything by N turns this into

$$[\dot{SS}] = r[SI] - p[SSI] + w[SI]. \tag{5}$$